

Lecture 2: Preferences and Utility

Alejandro Melo Ponce



Stony Brook University

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How to start modeling choices?

- Last lecture we had a buyer and a seller who faced a choice between buying and selling an object.
- However, there are usually more than two alternatives.
- So how to start modeling how individuals make choices?
- In most of economic theory, we will describe an economic agent only by his attitude towards the elements of some relevant set.
- We will assume that his attitude is expressed in the form of **preferences**.
- Today, we seek to come up to with a “proper” way to formalize the concept of preferences.
- After studying this concept, we will be concerned with when it is possible to represent preferences by “something handy”, called an **utility function**.

Preferences

- Although the goal is to construct a model of choice using preferences, we will examine first the concept by itself.
- Suppose that you want to fully describe the preferences of an agent towards the elements of a given set X .
- What conditions must the description fulfill? What conditions must the description include?
- Think back when you were applying to different universities before finding out you would come to Stony Brook.
- How did you describe your attitude towards different universities?
- Suppose you are at the ice-cream parlor. What is your attitude towards different flavors of ice-cream?

- Preferences should fully specify the attitude of the agent towards each pair of elements in X .
- For each pair, it should provide an answer to the question “How the agent compares the two alternatives?”
- I will present two equivalent ways to present this question, and formulate the required consistency conditions to make the responses “make sense”, i.e. to say that the responses are preferences.

The Questionnaire Q

- Let us think about preferences on a set X as **answers** to a long Questionnaire Q .
- Questions in Q consists of all questions of the type:

$Q(x, y)$ distinct x, y : How do you compare x and y ? Tick one and only one of the following three options:

- ☐ I prefer x to y (denoted $x \succ y$)
- ☐ I prefer y to x (denoted $y \succ x$)
- ☐ I am indifferent (denoted $x \sim y$)

- A legal answer to Q is a response in which **exactly only one of the boxes is ticked** in each question.

- By only allowing one of the three options above we exclude situations like:
 - ☐ They are incomparable.
 - ☐ I don't know what x is.
 - ☐ I have no opinion.
 - ☐ I prefer both x over y and y over x .
- And also dependence on other factors, like:
 - ☐ It depends on what my friends think.
 - ☐ It depends on the circumstances (sometimes I prefer x but sometimes I prefer y)
- And also we preclude intensity of preferences
 - ☐ I somewhat prefer x .
 - ☐ I love x and I hate y .

Consistency restrictions

- Not all legal answers to Q qualify as preferences over X .
- First we consider that the answer to $Q(x, y)$ must be identical to $Q(y, x)$.
- This is what is called a “framing effect”, in which people who are asked to compare two alternatives tend to prefer the first one. We exclude this possibility.
- We also require that the answer to $Q(x, y)$ and $Q(y, z)$ are consistent with the answer to $Q(x, z)$ in the following way: If x is preferred to y and y is preferred to z , then it must be the case that x is preferred to z . If the answers to $Q(x, y)$ and $Q(y, z)$ are indifference, then so is the answer to $Q(x, z)$.

Definition

Preferences on X are a function Q that assigns to any pair (x, y) of distinct elements exactly one of the three values $x \succ y$, $y \succ x$ or $x \sim y$ such that for any x, y, z in X , the following holds.

- No order effect $Q(x, y) = Q(y, x)$.
- **Transitivity:** if $x \succ y$, $y \succ z$ then $x \succ z$. If $x \sim y$, $y \sim z$, then $x \sim z$.

A discussion of transitivity

- How would you react if somebody told you that he prefers x to y , y to z but z to x ?
- What would you prefer?
 - “A weekend in Paris at a 4-star hotel with food quality Zagat 17 for \$574”,
 - OR
 - “A weekend in Paris for \$574 with food quality Zagat 17 at a 4-star hotel”.
- What if I ask you to compare each one of them to
 - “A weekend in Rome at a 5-star hotel with food quality Zagat 18 for \$612”

Systematic Violations of Transitivity

- Even though transitivity is an appealing concept, we are excluding individuals who base their judgements in procedures that cause a systematic violation of transitivity. Even though such procedures make sense.
- **Aggregation as a source of intransitivity:** An individual's attitude might be derived from the aggregation of more basic considerations.
- “Choosing an apartment” based on price, location and size.
- Suppose you have $X = \{A, B, C\}$ and you have that $A \succ_p B \succ_p C$, $B \succ_l C \succ_l A$, and $C \succ_s A \succ_s B$. You form your preferences between two apartments by pairwise majority.
- You will get A is preferred over B , B over C and C over A !

Systematic Violations of Transitivity

- **Similarities:** An individual might express indifference in a comparison between two elements that are too close to be distinguishable.
- Consider “I prefer Black over Light Gray”



- I perform a pairwise comparison of colors, going from Black, decreasing color saturation, until reaching Gray.
- Suppose I can't distinguish between two very close shades.
- $Black \succ LightGray$ but
 $Black \sim Gray1 \sim Gray2 \sim Gray3 \dots Gray100 \sim LightGray$.

Another way to think about preferences

- A second way to think about preferences is through an imaginary questionnaire R consisting of the following type of question:

$R(x, y)$ for all $x, y \in X$, not necessarily distinct. Is x as least as preferred as y ? Tick only one option

☐ Yes

☐ No

- A legal answer is
 - The answer to at least one of the questions $R(x, y)$ or $R(y, x)$ must be yes.
 - In particular, the silly question $R(x, x)$ must be yes.
 - For every x, y, z if the answer to $R(x, y)$ is yes and the answer to $R(y, z)$ is also yes then the answer to $R(x, z)$ should be yes.
- Whenever $R(x, y)$ has a “Yes” answer, we say $x \succsim y$

The traditional definition of preferences

Definition

Preferences on a set X is a relation $\succsim$ on X (a way to relate elements of X) satisfying:

- **Completeness:** For any $x, y \in X$ either $x \succsim y$ or $y \succsim x$.
- **Transitivity:** For any $x, y, z \in X$ such that $x \succsim y, y \succsim z$ then $x \succsim z$.

Proposition

Both definitions of preferences, either using the questionnaire Q or the questionnaire R are equivalent. A legal answer in Q constitutes a legal answer in R and the consistency requirements for answers in Q are equivalent to the consistency requirements for R

A response to: $Q(x, y)$ and $Q(y, x)$	A response to: $R(x, y)$ and $R(y, x)$
$x \succ y$	Yes and No
$x \sim y$	Yes and Yes
$y \succ x$	No and Yes

Utility functions and the utility representation

- Working with preferences is a bit cumbersome.
- In particular we would like to be able to apply some techniques from calculus for when we model consumer choice.
- So now we look at the questions of when can preferences be represented by a suitable index, and how this index would look like.
- In order to do this, we will work with the concept of utility.

Definition

A **utility function (index)** $u : X \rightarrow \mathbb{R}$ represents a preference relation $\succsim$ over X if and only if, for all $x, y \in X$ we have that $x \succsim y \Leftrightarrow u(x) \geq u(y)$.

- Is it always possible to find such a function?

The case for when X is a finite set

Proposition

Suppose X is finite. Then the assumptions of completeness and transitivity imply that there exist a utility function u that represents the preference relation $\succsim$ on X

- Define the set $B(x) = \{z \in X \mid x \succsim z\}$.
- Let $u(x) = \#B(x)$, i.e. the number of items on $B(x)$

Proof of the proposition

The argument has two parts:

- First part: Showing that $u(x) \geq u(y) \Rightarrow x \succsim y$.
- Second part: $x \succsim y \Rightarrow u(x) \geq u(y)$

What happens when X is infinite

- We need a technical definition to handle this case:

Definition

Let x_n and y_n be two sequences of elements of X . If $x_n \succsim y_n$ for every n and $x_n \rightarrow x$ and $y_n \rightarrow y$, then $x \succsim y$

- So we have in general:

Proposition

Every continuous preference relation can be represented by a continuous utility function.

- Remark: Utility is ordinal, only order matters: if u represents preferences, then for any increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(u(x))$ also represents the same preferences.

Indifference curves

- An indifference curve is the set of points that are indifferent between each other:

$$I(x) = \{z \in X | z \sim x\}$$

- With the aid of utility, we can express the same idea as follows:

$$I(x) = \{z \in X | u(z) = u(x)\}$$

Transitivity and indifference curves

